

Histograms and Cumulative Frequency

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

In a histogram where every class has the same width, what does the height of each bar represent?

.....

2 [2 marks]

Complete the cumulative frequency column.

Class	Frequency	Cumulative frequency
$0 < x \leq 10$	5	
$10 < x \leq 20$	9	
$20 < x \leq 30$	12	
$30 < x \leq 40$	4	

.....

3 [2 marks]

Using the completed table above,

(a) how many values are 30 or less?

(b) how many values are more than 20?

.....

4 [2 marks]

Using the same table (total frequency 30), write down the class interval that contains the median.

.....

5

[3 marks]

A frequency polygon is to be drawn for this table.

Class	0–10	10–20	20–30	30–40
Frequency	8	14	11	5

Write down the four points that should be plotted.

.....

6

[3 marks]

Work out the frequency density for each class.

Class width	10	10	10	10
Frequency	12	25	18	5

.....

7

[3 marks]

Class A is $0 < x \leq 10$ with frequency 20. Class B is $10 < x \leq 30$ with frequency 30.

- (a) Work out the frequency density of each class.
- (b) Which class has the values packed more closely together? Explain.

.....

8

[3 marks]

The table shows cumulative frequencies for the heights, h cm, of 50 plants.

Height	$h \leq 160$	$h \leq 170$	$h \leq 180$	$h \leq 190$
Cumulative frequency	12	30	46	50

Estimate how many plants are 175 cm or less.

.....

9

[4 marks]

The table shows cumulative frequencies for 60 values.

Value (x)	$x \leq 10$	$x \leq 20$	$x \leq 30$	$x \leq 40$	$x \leq 50$
Cumulative frequency	10	25	45	55	60

- (a) Estimate the median.
 (b) Estimate the lower quartile.
 (c) Estimate the interquartile range.

10

[4 marks]

A histogram has these bars.

Class	0–5	5–10	10–20	20–40
Frequency density	4	6	3	1

- (a) Work out the frequency of each class and the total.
 (b) Estimate how many values are greater than 15.

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A class is $20 < x \leq 50$ and its frequency is 45.
 Work out the frequency density.

2

[2 marks]

A histogram bar has frequency density 2.4 and covers a class of width 5.
 Work out the frequency of that class.

3

[3 marks]

Complete the frequency density column.

Class width	6	11	13	24
Frequency	18	33	26	12

4

[3 marks]

A class $20 < x \leq 40$ has frequency 36.

Estimate how many values lie between 25 and 40.

5

[4 marks]

A histogram has these bars.

Class	0–10	10–20	20–50	50–80
Frequency density	1.5	2.8	1.0	0.4

(a) Work out the total frequency.

(b) Estimate the median.

6

[4 marks]

The table shows cumulative frequencies for 80 values.

Value (x)	$x \leq 20$	$x \leq 40$	$x \leq 60$	$x \leq 80$	$x \leq 100$
Cumulative frequency	8	26	54	72	80

(a) Estimate the median.

(b) Estimate the quartiles.

(c) Work out the interquartile range.

7

[4 marks]

Using the same cumulative frequency table (80 values),

(a) estimate how many values are greater than 70.

(b) express that as a percentage of the total.

8

[4 marks]

Two groups of test marks have these summary values.

	Median	Lower quartile	Upper quartile
Group P	54	41	68
Group Q	60	54	67

(a) Work out the interquartile range for each group.

(b) Compare the two groups, giving two clear comparisons.

9

[4 marks]

A histogram has three bars. The first, for $0 < x \leq 20$, has frequency density 1.2. The second, for $20 < x \leq 30$, has frequency 30. The third covers $30 < x \leq 60$.

The total frequency is 90. Work out the frequency density of the third bar.

10

[5 marks]

A histogram of the times, t minutes, taken by 80 people has these bars.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 80$
Frequency density	1.2	2.8	1.4	0.3

(a) Work out the frequency of each class and show the total is 80.

(b) Estimate the median.

(c) Estimate how many people took more than 30 minutes.

(d) Estimate the interquartile range.

